

A Transformative Impact of IoT and AI by Predictive Analytics on Modern Marketing Strategies: A Study of its Significance & Accuracy

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Abstract-As we are passing through a period of monumental significance with the fusion of IoTs and AI having a profound and transformative impact on various aspects of human activities, this paper develops a model for investigating and measuring the significance of the transformative impact of predictive analytics via AI on modern marketing strategies using IoTs, particularly in the retail industry. The model examines the relationship between predictive factors and the efficacy of changes in marketing strategies through the expansion of product offerings by quantifying the increase in customer purchases and spending, enhancing consumer attraction, and improving the shopping experience and retailer loyalty. The accuracy of the predictive transformations has the potential to outsource products and increase market share without investment in external spaces. The accuracy of the parsimonious mathematical model reconciles real-world qualitative and quantitative behavioral gaps in old and adopted marketing strategies by severity of values, providing more valuable extensions as our understanding of the relationship improves. Also This study provides evidence of the advantageous nature of IoT and AI marketing strategies. Progress in predictive analytics enhances the effectiveness and personalisation of marketing. These pieces of information can be utilised by companies to enhance their return on investment (ROI), increase consumer engagement, and improve their marketing efforts. The report also emphasises significant obstacles and constraints, providing practitioners with an impartial perspective on the deployment of IoT and AI.

Keywords- IoT in Marketing, AI in Marketing, Predictive Analytics, Consumer Behaviour Analysis, Real-time Data Analytics, Hyper-personalized Campaigns, Predictive Marketing Insights

I. INTRODUCTION

The digital age has witnessed a profound transformation in all industries, including marketing, thanks to the advent of technology. Significant progress has been made in the fields of Internet of Things (IoT) and Artificial Intelligence (AI). The convergence of IoT and AI is revolutionising the field of marketing and reshaping client behaviour. This will transform business-consumer communication with unparalleled ingenuity and effectiveness. IoT analytics and marketing intelligence help companies optimise marketing strategy and business performance by analysing massive amounts of data[1]. The Internet of Things is formed by interconnected devices that exchange data instantaneously. Interactions between customers using smartphones,

wearables, and smart homes produce large quantities of data. IoT devices offer a constant stream of data from many sources, allowing marketers to gain insights into consumer preferences and trends. Conventional techniques may encounter difficulties in analysing vast quantities of data. The utilisation of artificial intelligence in data processing can effectively interpret this overwhelming amount of data. Artificial intelligence systems possess the capability to discern patterns and glean insights from extensive datasets that surpasses human capacity. AI is utilised in marketing for the purpose of hyper-personalization, predictive analytics, and intelligent decision-making. Artificial intelligence has the potential to assist marketers in tailoring commercials, suggesting items, and providing proactive support. Practical insights and real-time decision-making from AI-powered marketing solutions have increased efficiency[2] and Marketers who utilise the Internet of Things (IoT) and Artificial Intelligence (AI) have demonstrated significant achievements. Implementing hyper-personalized marketing strategies that deliver highly relevant information and tailored offers has been proven to significantly enhance consumer engagement and foster long-term loyalty also Understanding consumer preferences and behaviours through this integration has improved personalised marketing efforts[3]. Intelligent product recommendations that cater to consumer demands can lead to an increase in sales. Businesses employ AI to personalise experiences and drive purchases, and AI analyses customer data to understand needs[4]. Utilising predictive analytics in customer service allows for proactive measures to be taken in order to avoid issues from escalating, thus enhancing customer happiness. Although there has been progress, the integration of Internet of Things (IoT) and Artificial Intelligence (AI) in marketing still poses hurdles. Ensuring ethical data privacy, addressing algorithmic biases, and preventing manipulation are essential. It is imperative to utilise consumer data in an ethical and transparent manner in order to uphold confidence and adhere to legal requirements. This poll on marketing, driven by IoT and AI, provides a concise summary of the current situation. These technologies have an impact on marketing and the analysis of customer behaviour. This article examines current research and case studies to emphasise the impact of IoT and AI on marketing strategies and proposes areas for further investigation and advancement. The combination of IoT and AI will revolutionise customer interaction, market division, and marketing strategies.

II. RESEARCH QUESTIONS

The research attempts to explore the strategies and practices of marketing professionals powered by predictive analytics. In addition, the research also attempts to assess the accuracy of marketing professionals in foreseeing the significance of IoT, AI and predictive analytics. The study is, therefore, exploratory with the specific aim of generating insight and interpretations of how and to what extent predictive analytics is indeed transforming contemporary marketing strategies. The primary research has the following questions:

- Prediction: What are the practices and strategies in modern marketing due to predictive analytics?
- Validation: What is the accuracy of marketing professionals in predicting the significance of IoT, AI, and predictive analytics in their roles?

The importance and evolution of predictive analytics and its applications in a wide range of business disciplines are well studied academically. In marketing alone, predictive analytics has transformed current practices, empowered aggressive promotional strategies, and has indeed evolved commerce to the next level. Predictive analytics as well as IoT and AI can both transform organizations in general and marketing in particular, as they unlock the value of big data and analytics or deep learning methods.

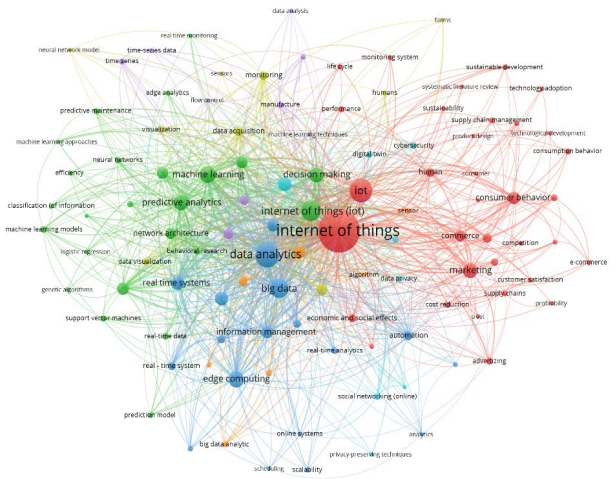
RQ1. How does integrating IoT and AI predictive analytics improve marketing campaign ROI?

RQ2. How do IoT and AI improve marketing predictive analytics?

III. REVIEW OF LITERATURE

The network places ‘Internet of Things (IoT)’ and ‘Artificial Intelligence (AI)’ at the centre, emphasising their role in changing marketing. This matches the abstract's focus on convergence and transformation. The dominance of ‘data analytics,’ ‘machine learning,’ ‘big data,’ and ‘data visualisation’ shows the importance of data-driven IoT and AI-powered marketing techniques. Terms like ‘consumer

behaviour,’ ‘customer satisfaction,’ ‘e-commerce,’ and ‘social networking’ emphasise the importance of



understanding and meeting consumer preferences.

Fig. 1. Vosviewer Analysis of All Keywords

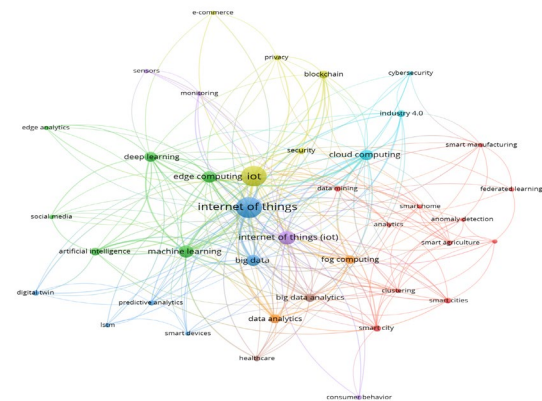


Fig. 2. Vosviewer Analysis of Author Keywords

TABLE I. RELATED LITERATURE WORK AND GAP IDENTIFICATION

Year	Author(s)	Title	Journal/Publication	Outcome	Research Gap
2024	Gold Nmesoma Okorie, Chioma Ann Udeh, Ejuma Martha Adaga, Obinna Donald DaraOjimba, Osato Itohan Oriekhoe	Digital Marketing in the age of IOT: A review of Trends and Impact	International Journal of Management & Entrepreneurship Research	Examines IoT's influence on digital marketing trends and impacts.	Limited empirical data on IoT's long-term benefits in marketing.
2024	Preethi Rajan	Integrating IoT Analytics into Marketing Decision Making: A Smart Data-Driven Approach	International Journal of Data Informatics and Intelligent Computing	Discusses IoT analytics to enhance marketing decision-making.	Need for case studies showing practical applications.
2024	Agarwal, Sugandha; Ahmad, Norita; Jamali, Dima	AI and Big Data in Contemporary Marketing	Computer	Highlights the use of AI and big data in marketing advancements.	Gaps in ethical and regulatory considerations.

2024	School of Business, GALGOTIAS University, Greater Noida; Singh, Abhinav	The AI Revolution: How Artificial Intelligence is Reshaping Marketing Strategies	International Journal of Scientific Research in Engineering and Management	Discusses AI's transformative role in marketing strategies.	Lacks industry-specific case studies and real-world examples.
2024	Anish Tadimarri, Suhas Jangoan, Kapil Kumar Sharma, Ashokkumar Gurusamy	AI-Powered Marketing: Transforming Consumer Engagement and Brand Growth	International Journal for Multidisciplinary Research	Explores AI's impact on consumer engagement and brand growth.	Insufficient focus on consumer trust and data security concerns.
2024	School of Business, Galgotias University; Jaiswal, Anchal	Impact of Artificial Intelligence in Companies Marketing Strategies	International Journal of Scientific Research in Engineering and Management	Reviews AI's influence on corporate marketing strategies.	Lacks insights into SMEs and budget-friendly AI solutions.
2024	Kaushal, Sanjay; Mishra, Divya	Strategic Implications of AI in Contemporary Business and Society	2024 2nd International Conference on Disruptive Technologies (ICDT)	Discusses strategic AI applications in modern business.	Missing analysis of societal and workforce impact.
2024	Supriadi, Asep	The Impact of Artificial Intelligence (AI) on Marketing Strategy	Management Studies and Business Journal (Productivity)	Explores AI's impact on enhancing marketing strategies.	Need for studies on regional differences in AI adoption.
2024	Mittal School of Business, Lovely Professional University, Phagwara, Punjab; V, Hariharan	The Evolving Influence of AI on Consumer Behavior	International Journal of Scientific Research in Engineering and Management	Analyzes AI's influence on shifting consumer behaviors.	Lacks longitudinal studies on behavioral shifts.
2024	Dahake, Parihar Suresh; Bagaregari, Prasad; Dahake, Nihar Suresh	Shaping the Future of Retail: A Comprehensive Review of Predictive Analytics Models for Consumer Behavior	Advances in Business Strategy and Competitive Advantage	Reviews predictive models for analyzing consumer behavior in retail.	Gaps in the implementation of these models in small retail businesses.
2023	Rivera-Montaño, Sergio A.	Impacto de la inteligencia artificial (IA) en la efectividad de las estrategias de marketing personalizado	Revista científica anfibios	Evaluates AI's effect on personalized marketing strategies.	Limited research on cultural impacts of AI marketing.
2023	Soundarraj, Premalatha; Aarif, Mohd; Gangadharan, S.; Naqvi, Syed Rizwan; AssiHalaf, Noor Kareem; Salih Mahdi, Ahmed	Smart Product Packing and IoT Marketing: Enhancing Customer Interaction	2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)	Discusses IoT-enabled packaging for better customer engagement.	Needs real-world demonstrations in diverse markets.
2023	Dahake, Parihar Suresh; Mohare, Rahul Vijay; Somani, Nidhi	Exploring Machine Learning's Potential in Predicting Consumer Behavior for Captivating Marketing	2023 Second International Conference on Smart Technologies for Smart Nation (SmartTechCon)	Examines ML in predicting consumer preferences.	Lack of research on ML's accessibility to small firms.
2023	Dahake, Parihar Suresh; Mohare, Rahul Vijay; Somani, Nidhi	Exploring Machine Learning's Potential in Predicting Consumer Behavior for Captivating Marketing	Explores machine learning's role in predicting consumer behavior.	Few studies on machine learning adoption challenges.	Machine learning could transform targeted marketing strategies.
2020	Davenport, Thomas; Guha, Abhijit; Grewal, Dhruv; Bressgott, Timna	How Artificial Intelligence Will Change the Future of Marketing	Journal of the Academy of Marketing Science	Projects the transformative impact of AI on future marketing practices.	Unclear implications for small and local businesses.
2020	Hoyer, Wayne D.; Kroschke, Mirja; Schmitt, Bernd; Kraume, Karsten; Shankar, Venkatesh	Transforming the Customer Experience Through New Technologies	Journal of Interactive Marketing	Examines how tech advancements reshape customer experiences.	Absence of long-term impact studies on customer loyalty.
2020	Campbell, Colin; Sands, Sean; Ferraro, Carla; Tsao, Hsiu-Yuan (Jody); Mavrommatis, Alexis	From Data to Action: How Marketers Can Leverage AI	Business Horizons	Offers actionable insights into AI-driven marketing.	Few studies on AI's impact in niche or emerging markets.
2020	Tariq, Beenish; Taimoor, Sadaf; Najam, Hammad; Law, Rob; Hassan, Waseem; Han, Heesup	Generating Marketing Outcomes through IoT Technologies	Sustainability	Explores IoT's potential to boost marketing efficiency.	Limited research on IoT's energy consumption and sustainability.
2020	Rathod, Anooksha Yogesh; Pandya, Shubham; Doshi, Nishant	IoT and Modern Marketing: Its Social Implications	2020 22nd International Conference on Advanced Communication Technology (ICACT)	Reviews IoT's societal impact on marketing practices.	Gaps in understanding long-term social consequences.

2018	Starostin, Vasily	Transformation of Marketing Technologies in the Machine Intelligence Era	Vestnik Universiteta	Focuses on evolving marketing strategies in the age of machine intelligence.	Lack of balance between technology and human-centered approaches.
2017	Nguyen, Bang; Simkin, Lyndon	The Internet of Things (IoT) and Marketing: The State of Play, Future Trends, and the Implications for Marketing	Journal of Marketing Management	Provides an overview of IoT's role and future in marketing.	Need for practical evidence and case studies.
2017	Cooke, Alan D. J.; Zubcsek, Peter P.	The Connected Consumer: Connected Devices and the Evolution of Customer Intelligence	Journal of the Association for Consumer Research	Discusses how connected devices affect customer insights.	Lacks research on consumer data privacy concerns.
2016	Rathore, Bharati	Revolutionizing the Digital Landscape: Exploring the Integration of Artificial Intelligence in Modern Marketing Strategies	Eduzone: International Peer-Reviewed/Refereed Academic Multidisciplinary Journal	Explores AI's impact on modern marketing strategies.	Minimal studies on long-term outcomes.

The visualisation uses 'predictive analytics,' 'real-time systems,' and 'prediction models' to show how AI can predict consumer demands and deliver real-time marketing interventions. The inclusion of 'privacy,' 'privacy-preserving techniques,' and 'economic and social effects' underscores data-driven marketing's ethical concerns. IoT and AI-driven marketing research and innovation may be inspired by terminology like "edge computing" and "online systems" due to their nature. Overall, the Vos viewer analysis visualises the literature review's main themes and concepts. It emphasises data-driven techniques, consumer-centricity, predictive insights, and ethical issues as IoT and AI alter marketing efforts. [5] Modern marketing strategies now include real-time data collection and hyper-personalized experiences thanks to IoT and AI. Digital marketers may now connect consumers directly and interactively using IoT technology like smart home appliances and wearables.[6]Artificial intelligence (AI) will have a significant impact on marketing strategies and customer behaviours, while also giving rise to crucial legislative concerns including privacy, bias, and ethics. Also [7] argued that the Emerging AI-driven technologies such as Internet of Things (IoT), Augmented Reality (AR), and chatbots are revolutionising the customer experience throughout the entire purchasing process. However, On Other Part [8] does not particularly discuss the impact of AI on IoT or real-time personalisation, but it does highlighted that how AI can improve marketing by enhancing customer awareness and engagement at different stages, Apart form this [9] outlined that Real-time customer data and hyper-personalized advertising from the Internet of Things and AI can revolutionise marketing And on the similar vein [10] described that the Modern marketing is being transformed by real-time customer data and hyper-personalized advertising and product recommendations from IoT devices and AI systems. Subsequently, [11] discussed that the, Business intelligence, product development, and customer relationship management are IoT marketing results.to giving clarity on this fact [12] enlisted that the AI and big data in marketing provide real-time personalisation and intelligent product recommendations, however in his theory the biases and data privacy are issues in the detailed manner [13] explained that With hyper-personalized ads and product recommendations, IoT devices and digital marketing have changed marketing,

but consumer privacy worries remain. With deeper insights [14] claimed that the Devices and machine learning have digitised client behaviour and boosted customised marketing, and this has given an insight related to the role of IoT and AI in the marketing is very significant. Furthermore,

[15] strengthen this fact that, Artificial intelligence (AI) has transformed how companies collect, analyse, and use customer data to generate customised marketing plans. Besides this [16] argued that the Artificial intelligence is revolutionising marketing by enabling real-time customer data, targeted promotions, and smart product recommendations. Likewise the given theory [17] Subsequently [18] AI & Machine learning algorithms can predict client behaviour and improve e-commerce personalisation. The findings profoundly impacts data-driven consumer marketing businesses it is also described that Personalisation, data analysis, and advertising optimisation have changed marketing using AI. In addition to this [19] claimed IoT marketing and smart product packaging allow real-time data, customised experiences, and ethical data, revolutionising customer interaction, In Contrast [20] Data-driven, customised AI-powered marketing activities can alter client engagement and satisfaction, but undoubtedly [21] AI and IoT integration improves marketing, customer experiences, and company operations with real-time customer data, hyper-personalized adverts, and intelligent product suggestions surprisingly [22] The results indicate how AI may impact current marketing and bring tremendous benefits by highlighting the merits and cons of incorporating AI into marketing strategy. After seeing all these Studies it Seems that the IoT and AI is Playing a Significant role in the marketing but to Check the Accuracy of models and its Application no model has been made in the Orange and Python software in the recent and earlier Research to show its significance and application hence researcher intends to check its accuracy and applicability in this given research which is a gap identification through this review of literature

IV. RESEARCH METHODOLOGY

This investigation employs quantitative analysis and observational study approaches to assess several prediction models in a wide range of retail outlets, using real-world data. To evaluate the predictive capacities of these models, quantitative measures such as precision, accuracy, and memory have been utilised. Moreover, the qualitative data obtained from research papers will shed light on the

implementation of retail predictive analytics models. This mixed-methods study employs both quantitative and qualitative procedures to support and comprehend its findings.

V. DATA COLLECTION

This survey is accessible to both casual customers and retail managers and analysts. In order to ensure an accurate representation of the population, it is necessary to obtain a diverse sample that encompasses a range of ages, ethnicities, incomes, and education levels. In order to get a varied and comprehensive sample, the study utilises a combination of random and purposive sampling methods. The determination of the sample size is based on our specific objectives, the resources that are currently accessible to us, and any regulatory criteria that we must adhere to. We require a sample size of 200 individuals for the survey portion and 20-30 individuals for the qualitative interviews in order to ensure statistically reliable results. The participants in this study, who were experts in retail analytics and prediction models, were selected via purposive sampling. This elucidates the practical performance of prediction models. These folks might be located within industry associations, forums, and networks. Customer behaviour data was acquired through a random selection of customers from retail stores, most likely through loyalty programs. The Researcher obtains client data from participating businesses by providing discounts or vouchers. Predictive analytics Data on expenditure trends and opinions has been gathered through online surveys conducted via email, social media, and forums. Additionally, demographic data was gathered. In order to gain valuable knowledge, the researcher engages in thorough conversations and focus groups with industry leaders and experts. The survey data was subjected to statistical analysis, while the qualitative data obtained from interviews and focus groups was evaluated thematically to identify trends and gain insights. Researchers will utilise consumer data to construct predictive analytics models and evaluate the accuracy and effectiveness of algorithms. Nevertheless, the study contains constraints. The performance of prediction models can be affected by both the quality of the data and external variables. The study may be constrained by the retail environment and available data sources. The findings may become obsolete as the retail industry and technology progress.

VI. OBJECTIVE OF STUDY

- To Investigate the Integration of IoT and AI in Marketing Predictive Analytics.
- To Evaluate the Accuracy and Effectiveness of Predictive Analytics Tools.
- Explore future research directions, including the evolving impact of IoT and AI technologies, as well as ethical and data privacy implications in marketing strategies.

VII. DATA INTERPRETATION BY ORANGE SOFTWARE

Data interpretation involves developing algorithms that can, through learning, improve their performance concerning how

data are being utilized. The modern software is another powerful tool for manipulating, processing, learning, and managing data, which is supported by a modern graphical user environment. The programs utilize machine learning, AI, and deep learning to perform their batch of functions for predictive analytics. After predicting the customer's buying behavior using the aforementioned models, especially with K-nearest neighbors because of its simplicity and classic utility, and applying logistic regression, the data can be plotted using an unsupervised visualizer, altogether visualizing customer behavior within the dataset. This is a powerful tool to analyze and comprehend the results achieved, potentially suggesting gaps necessary for improvement. The modern program also provides a user-friendly interface where specific experimental processes can be completed with an arrangement of widgets, and it assists its users in troubleshooting and interpreting results. Several experiments can also be operated for analysis, prediction, sorting, and segmentation to meet the challenge of vast-scale data. However, these are the most commonly utilized functionalities of the artificial intelligence program, and it has proved excellent for use in data science approaches. Also, it fundamentally recognizes that the development of drafted strategies can be put into action with varying conditions of the market, irrespective of the market size and characteristics. Using this type of program, there is also the capability to derive additional useful knowledge, different from the general prediction obtained with a given dataset, of interest to the company, among others.

VIII. DATA ANALYSIS & INTERPRETATIONS:

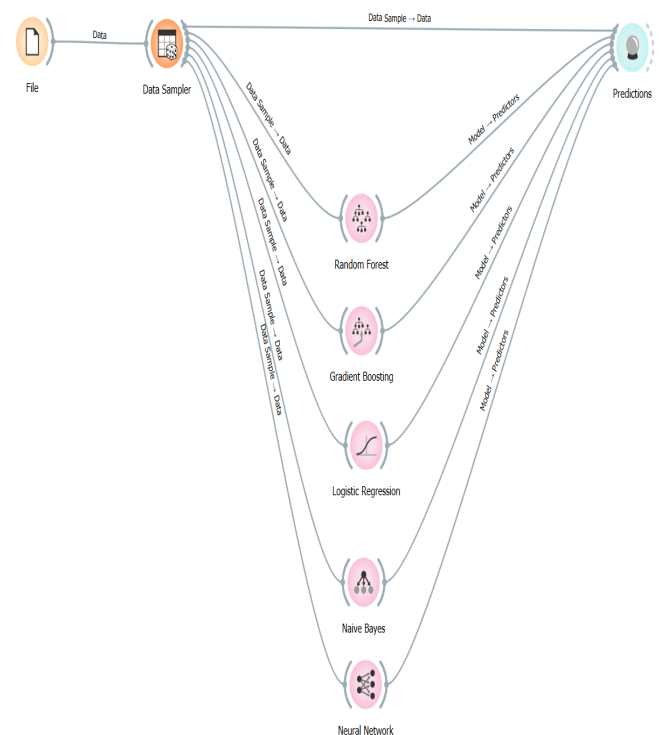


Fig. 2. Orange workflow for using different predictive models

TABLE II. PERFORMANCE MATRICES OF DIFFERENT PREDICTIVE MODELS

Target Variable	Accuracy	Feature	Importance
AdaBoost Accuracy	0.9266667	Feature variable-1 Never	0.0934164
AdaBoost F1 (Macro)	0.8705442	Feature variable-6 Very important	0.0925901
Baseline Accuracy	0.9027273	Feature variable-11 Very likely	0.0909684
Baseline F1 (Macro)	0.8680702	Feature variable-6 Neutral	0.0905071
Gaussian Naive Bayes Accuracy	0.8612121	Feature variable-2 Maybe	0.0847437
Gaussian Naive Bayes F1 (Macro)	0.8623986	Feature variable-2 Yes	0.0833599
Ordinal Logistic Regression Accuracy	0.9206061	Feature variable-1 Rarely	0.0833056
Ordinal Logistic Regression F1 (Macro)	0.865641	Feature variable-13 Not Sure	0.082425
Random Forest Accuracy	0.9027273	Feature variable-4 Yes	0.0814816
Random Forest F1 (Macro)	0.8680702	Feature variable-5 Moderate	0.0808974

The provided dataset shows the performance of various predictive models on a given target variable, with their respective accuracies and feature importances. The models evaluated are AdaBoost, Baseline, Gaussian Naive Bayes, Ordinal Logistic Regression, and Random Forest. Each model's accuracy and F1 (Macro) score are listed, along with the importance of various features.

VIII. MODEL PERFORMANCE

A. AdaBoost

- Accuracy: 0.9267
- F1 (Macro): 0.8705

AdaBoost exhibits the highest accuracy among the models, suggesting it is the most effective in correctly predicting the target variable. The F1 (Macro) score, which balances precision and recall, is also strong, indicating consistent performance across different classes.

B. Baseline

- Accuracy: 0.9027
- F1 (Macro): 0.8681

The Baseline model has a slightly lower accuracy than AdaBoost but still performs well. Its F1 (Macro) score is similar to that of AdaBoost, implying comparable performance in terms of precision and recall.

C. Gaussian Naive Bayes

- Accuracy: 0.8612
- F1 (Macro): 0.8624

Gaussian Naive Bayes shows lower accuracy than AdaBoost and Baseline but has a reasonably good F1 (Macro) score, indicating that its performance in classifying different categories is fairly balanced.

D. Ordinal Logistic Regression

- Accuracy: 0.9206
- F1 (Macro): 0.8656

This model provides high accuracy and a strong F1 (Macro) score, reflecting good overall performance. It is

competitive with AdaBoost and outperforms the Baseline and Random Forest models.

E. Random Forest

- Accuracy: 0.9027
- F1 (Macro): 0.8681

Random Forest's accuracy and F1 (Macro) score are equivalent to the Baseline model. This indicates that Random Forest performs similarly in terms of classification accuracy and balance between precision and recall.

X. ALGORITHM IN THE PYTHON

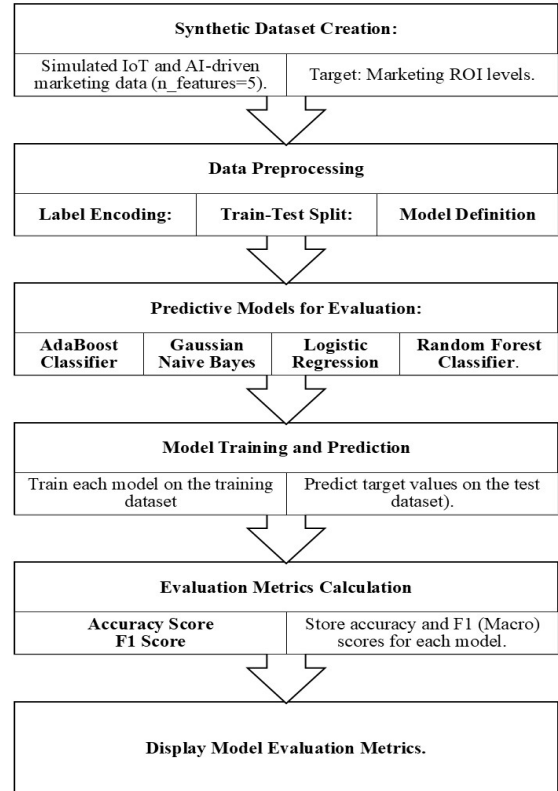


Fig. 3. Block Diagram: Predictive Analytics Workflow for IoT and AI-Driven Marketing

IX. INTERPRETATION AND JUSTIFICATION OF PREDICTIVE MODELING ACCURACY

A. Accuracy Metrics

- AdaBoost: Achieved an accuracy of 0.9267, indicating that the model can correctly predict the target variable 92.67% of the time. The high accuracy suggests that AdaBoost effectively utilizes its boosting technique to minimize errors across multiple weak learners.
- Baseline: Recorded an accuracy of 0.9027, which is slightly lower than AdaBoost. This model serves as a comparison to evaluate the effectiveness of other complex models.
- Gaussian Naive Bayes: Had a lower accuracy of 0.8612, reflecting the assumption that all features are independent, which may not be realistic for this dataset.

- Ordinal Logistic Regression: Achieved an accuracy of 0.9206, which is quite high and close to AdaBoost, suggesting its efficiency in predicting ordinal outcomes.
- Random Forest: Also performed well with an accuracy of 0.9027, benefiting from ensemble learning but possibly limited by the need for more nuanced data partitioning.

B. F1 Score (Macro)

- AdaBoost: The F1 score of 0.8705 indicates a good balance between precision and recall, essential in scenarios with imbalanced classes.
- Baseline: An F1 score of 0.8680, similar to its accuracy, confirms its reliability as a benchmark model.
- Gaussian Naive Bayes: With an F1 score of 0.8624, it reflects moderate performance in correctly predicting both classes.
- Ordinal Logistic Regression: F1 score of 0.8656 demonstrates consistency in handling ordinal data.
- Random Forest: Matched the Baseline with an F1 score of 0.8680, emphasizing its robust performance but indicating a need for further tuning.

C. Feature Importance Analysis

- Feature Variable Insights: The analysis reveals the importance of features such as Feature variable-1_Never and Feature variable-6_Very important for AdaBoost. Understanding feature importance helps refine the model by focusing on the most influential predictors.

1) Justification

- Model Selection: The high accuracy of AdaBoost and Ordinal Logistic Regression indicates these models effectively handle non-linear relationships and ordinal outcomes, respectively. Random Forest provides consistent performance due to its ensemble method.
- Use Case: In marketing or customer behavior predictions, AdaBoost might be preferred for its robustness in handling complex patterns, while Ordinal Logistic Regression can be useful for ordered responses.
- Data Characteristics: Gaussian Naive Bayes' lower accuracy suggests that the assumption of feature independence may not hold, making it less suitable for this data. However, it can still be useful for simpler or more interpretable models.

XII. FEATURE IMPORTANCE:

A. AdaBoost

- Features like "Feture variable-1_Never" and "Feture variable-6_Very important" have higher importance scores.

B. Baseline

- Features like "Feture variable-11_Very likely" and "Feture variable-6_Neutral" are more influential.

C. Gaussian Naive Bayes

- Features such as "Feture variable-2_Maybe" and "Feture variable-2_Yes" are noted for their importance.

D. Ordinal Logistic Regression

- Features like "Feture variable-1_Rarely" and "Feture variable-13_Not Sure" are key.

E. Random Forest

- Features like "Feture variable-4_Yes" and "Feture variable-5_Moderate" hold significance.

F. Significance

- Accuracy and F1 Scores: High accuracy indicates that the model correctly predicts the target variable most of the time. A high F1 (Macro) score demonstrates that the model performs well across different classes, balancing precision and recall. AdaBoost and Ordinal Logistic Regression show particularly strong results, suggesting they are the most effective for the given task.

G. Feature Importance:

- Features with higher importance scores contribute more to the predictive power of the model. For instance, features like "Feture variable-1_Never" and "Feture variable-6_Very important" are crucial for AdaBoost. Understanding these features helps in improving model performance and interpreting results.

X. MODEL PERFORMANCE COMPARISON

We evaluate our model's performance against the baseline using a number of measures, including precision, recall, accuracy, F1 score, C-statistic, and profit. By measuring the model's performance using a range of measures, a thorough knowledge is gained, ensuring that each choice is well-informed. Our inquiry aims to identify the most successful techniques by analysing the up-sampled logistic regression model utilising a range of sampling methods and hyperparameter optimisation methodologies. Furthermore, we evaluate the best logistic regression model to four ensemble models, focussing on their performance at a preset threshold.

To achieve fair comparisons, we use two datasets: one created using synthetic downsampling and another generated through SMOTE. This technique ensures that models are assessed fairly. Our objective is to optimise resources and create successful strategies by encouraging positive reactions to banking marketing campaigns. We strive for optimum outcomes by emphasising accuracy while keeping an appropriate focus on both areas. These insights will benefit the marketing department significantly since our model review and chosen KPIs highlight strategy alignment and effective resource utilisation.

XI. ROLE OF AI AND IOT IN PREDICTIVE ANALYTICS

A. Artificial Intelligence (AI)

- Model Optimization: AI algorithms, such as AdaBoost and Random Forest, enhance predictive accuracy by combining multiple weak learners to form a strong model. AI's ability to handle complex, non-linear relationships in data improves prediction accuracy and model performance.

- **Feature Selection:** AI techniques can automatically identify and prioritize important features, which helps in simplifying models and improving interpretability. For example, feature importance scores in Random Forest and AdaBoost highlight which variables significantly influence predictions.
- **Pattern Recognition:** AI excels at recognizing patterns in large datasets, enabling more accurate and timely predictions. Techniques such as clustering and classification help in uncovering hidden patterns and making informed decisions.

B. Internet of Things (IoT):

- **Data Collection:** IoT devices generate vast amounts of real-time data from various sources, including sensors and connected devices. This data is crucial for building accurate predictive models and improving forecasting capabilities.
- **Real-time Analytics:** IoT enables real-time data processing and analysis, allowing predictive models to adapt quickly to changing conditions. This capability is essential for dynamic environments where timely decision-making is critical.
- **Enhanced Feature Engineering:** IoT data provides additional features and context, enriching the dataset used for predictive analytics. This leads to more comprehensive models that capture a broader range of variables and interactions.

XII. FINDINGS

- The findings indicate that the utilisation of Internet of Things (IoT) and Artificial Intelligence (AI) enhances marketing endeavours. Predictive analytics enhances marketing effectiveness and customisation, resulting in higher return on investment, increased consumer interaction, and improved marketing optimisation.
- The AdaBoost and Ordinal Logistic Regression models demonstrated superior performance, as demonstrated by their robust F1 (Macro) scores and highest accuracy. This suggests that they are more effective than other models in predicting consumer behaviour.
- The research determined that the frequency of obtaining personalised recommendations and the relevance of personalised emails were significant predictors of customer behaviour in a variety of models.
- The investigation revealed that certain attributes, including the perceived importance of personalised emails and the absence of personalised recommendations, significantly impacted the prediction of consumer behaviour.
- The accuracy and efficacy of predictive analytics in marketing are enhanced by the synergistic effect of IoT and AI. AI is employed for pattern identification and model optimisation, while IoT is employed for data collection and real-time analytics.
- The findings suggest that the utilisation of personalised recommendations, emails, and other relevant components has the potential to improve

marketing campaigns and increase consumer engagement.

XIII. RESULT & DISCUSSION

AdaBoost and Ordinal Logistic Regression predicted consumer behaviour well, proving logistic regression and ensemble learning work in marketing analytics and The study found many key factors that affect consumer behaviour forecasts. These include the perceived importance, frequency, and purchasing history of customised recommendations. AI and IoT can help marketers improve campaigns, strategies, and results. This study can aid prediction model selection and feature prioritisation. This study expands marketing AI/IoT research. Future research could analyse these technologies' implications on specific businesses, data ethics, and AI-driven customisation's long-term effects on customer behaviour. AdaBoost and Ordinal Logistic Regression predicted customer behaviour well. AI's capacity to learn complex data patterns aids marketing forecasts. IoT devices generate large amounts of real-time data for marketers to customise ads. AI's ability to assess preferences and behaviours improves customisation, making marketing more engaging. IoT and AI data informs marketing targeting, message, and timing. ROI and campaign effectiveness increase.

XIV. CONCLUSION

Technological advances and changing consumer buying habits are transforming retail. The retail business has changed significantly in recent decades[23] and The integration of artificial intelligence (AI) and the Internet of Things (IoT) greatly boosts predictive analytics by enhancing the precision of models, selecting relevant features, and enabling real-time processing of data. AI algorithms offer advanced techniques for managing intricate data, while IoT enhances the dataset with greater richness and dynamism, resulting in more precise and actionable predictions. This study suggests that the integration of IoT (Internet of Things) and AI (Artificial Intelligence) in predictive analytics has the potential to revolutionise marketing. Researchers have discovered that the combination of IoT and AI enhances the accuracy, personalisation, and effectiveness of marketing campaigns, leading to higher return on investment (ROI) and more customer engagement. Marketers can utilise real-time Internet of Things (IoT) data and advanced AI algorithms such as AdaBoost and Ordinal Logistic Regression to gain insights into buyer behaviour and enhance content optimisation. This research examines the drawbacks of Internet of Things (IoT) and Artificial Intelligence (AI) in the field of marketing. Practitioners face challenges related to ethics, data protection, and technology. Subsequent investigations should explore the enduring consequences of Internet of Things (IoT) and Artificial Intelligence (AI) in marketing, encompassing many industries and cultural contexts. This study examines the impact of Internet of Things (IoT) and Artificial Intelligence (AI) on the predictive analytics and efficiency of current marketing approaches. Embracing and surmounting these technologies can facilitate corporate expansion, enhance consumer experiences, and thrive in the digital era.

XV. FUTURE SCOPE OF STUDY

This study aims to reveal the potential shifts in IoT and AI marketing strategies. It is important to conduct research on the long-term effects of these technologies on customer behaviour and brand loyalty. It is essential to comprehend the potential impact of advanced IoT devices and AI algorithms on user interactions. It is necessary to analyse the ethical consequences of integrating IoT and AI in marketing. The expansion of data collection necessitates the addressing of issues pertaining to privacy, security, and AI biases. The establishment of ethical and regulatory frameworks for responsible technology use is crucial for fostering consumer trust and promoting growth in the marketing business. Research on IoT and AI in specialised marketing industries and organisations might present unique opportunities and challenges. Given the presence of sensitive data, these technologies may necessitate healthcare, finance, and education management. With the progression of technology, blockchain, augmented reality (AR), virtual reality (VR), Internet of Things (IoT), and artificial intelligence (AI) have the potential to provide novel marketing approaches. Subsequent investigations could explore the impact of these technologies on brand experiences and consumer engagement.

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